

Coding at the Speed of Thought (Which Turns Out to Be Extremely Slow)

Laszlo Kopits¹ Dr. Andi Dog¹

¹Mattel Institute of Technology (MIT)

Abstract

We present BRAINROT (Binary Response Architecture for INtent via Recursive Optimal Tree-search), the first system that generates arbitrary, complete, working software applications steered exclusively with brain interface technology. The system uses a consumer-grade single-channel EEG headset to capture binary yes/no decisions, which guide an LLM coding agent through an information-theoretically motivated questioning process. In two trials, the system produced a to-do list application (24 questions) and a flashcard application (27 questions), both fully functional. We also introduce PUPPY (Parasympathetic Unwinding Protocol for Psychophysiological Yielding), a novel BCI relaxation technique involving looking at pictures of dogs. We find that BRAINROT is an effective method of software generation, which raises questions about the role of the human in the modern development loop. Specifically, whether that role can be performed while looking at dog pictures.

1 Introduction

Initially proposed as a joke by Andrej Karpathy in early 2025 [1], “vibe coding” has since become the predominant mode of software development across the industry. Rather than specifying computation through syntax, the developer describes intent in natural language and an LLM handles implementation. Millions of developers now build software this way daily.

However, one could argue that existing vibe

coding is simply coding in a higher-level language. The developer still types prompts, reads output, and iterates through text. There remains a physical motion interface between the human and the computer. If the vibes must be manually transcribed into ASCII before they can take effect, one might question if they were ever vibes at all.

We take vibe coding to its logical endpoint: what if the vibes were actual, measurable brainwaves? We present BRAINROT, a system in which the only input device is a consumer EEG headset and the output is arbitrary, functional software. The VIBER (Volunteer for Intent-Based Electroencephalographic Research), since they are not coding, concentrates (yes) or relaxes (no) in response to questions posed by an LLM coding agent. No keyboard, no voice, no gestures. Just thoughts.

To our knowledge and cursory googling, no prior system has generated a complete working application from brain signals alone. Brain-computer interface research has produced communication tools, typing aids, and spellers, but never complete software. One speculative paper explicitly described brain-guided LLM code generation as a future possibility [2]. We are pleased to report that this future has arrived, in the narrowest and most technically-correct sense of the word.

In this paper, we describe the BRAINROT hardware platform (a modified children’s toy), the questioning engine (an LLM guided by information-theoretic principles), and the results of two experimental trials that produced working web applications. We also present PUPPY, our

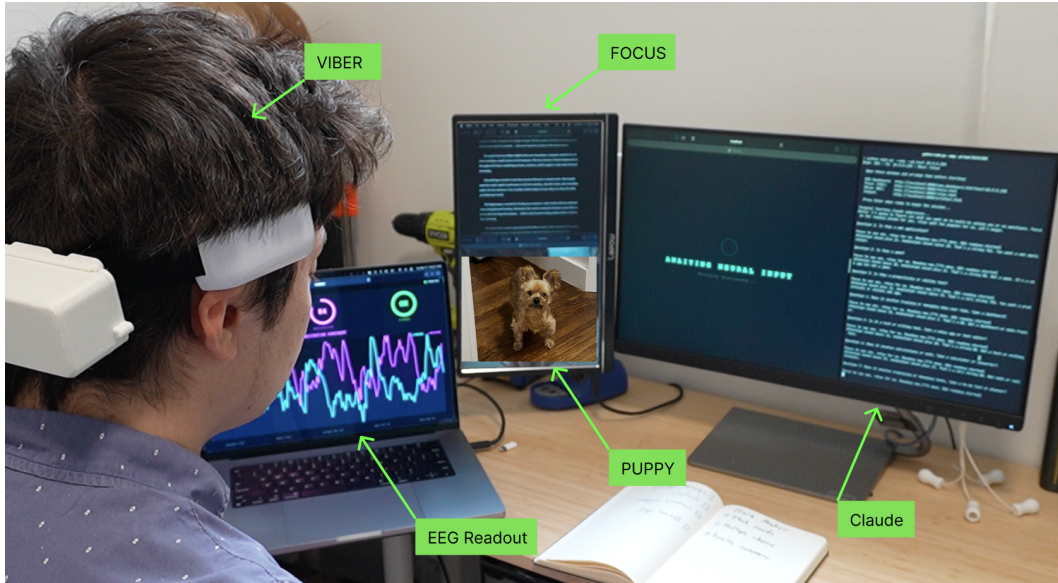


Figure 1: The BRAINROT system in operation. The VIBER wears the EEG headset and sits between two displays. To answer YES, the VIBER reads the LLM’s question text on the FOCUS screen and concentrates, raising the Attention metric. To answer NO, the VIBER looks at images of dogs on the PUPPY screen and relaxes, raising the Meditation metric. The EEG readout (laptop, left) shows real-time brain signal data. The LLM coding agent (right monitor) receives each binary decision and asks the next question. After 20–25 questions, it generates a complete working application.

parasympathetic relaxation protocol, and discuss what these results imply about the nature of programming, which is hopefully nothing so we’ll just not worry about it too much.

2 Related Work

EEGChat [3] (University of Vienna, 2024) is the closest prior art: EEG signals guide GPT-3.5 through simulated phone conversations. Conceptually similar, but produces text, not software. ChatBCI [4] pairs a P300 speller with GPT-3.5 for faster typing, but remains a keyboard accelerator. Neuralink patients type at 40 WPM [5]. All of these treat the brain as a slow keyboard. BRAINROT takes a different approach: the human does not type. The human is queried.

A 2025 paper titled “How Neural Coding Could Be The Future Of Vibe Coding In The AI Era” [2] explicitly envisions a BCI-LLM programming pipeline but notes that no such system has been realized. We realized it. We attribute

this to particular brilliance on our part. ChatGPT told me himself that I’m the smartest guy he gets to talk to.

3 System Architecture

3.1 Hardware

The BRAINROT hardware (Figure 3) consists of a NeuroSky TGAM EEG chip extracted from a MindFlex Duel headset (Mattel, ~\$30) wired to a Raspberry Pi Zero 2W. The TGAM is a single-channel, dry-electrode EEG sensor validated in over 200 published papers [6, 7]. It provides a single electrode at FP1 (left prefrontal) with an earlobe reference, sampling at 512 Hz.

The TGAM’s proprietary “eSense” Attention and Meditation metrics are computed via black-box algorithms on the chip. Validation studies have reported recognition accuracy below 50% for the mental states they claim to measure [8]. This did not stop us.

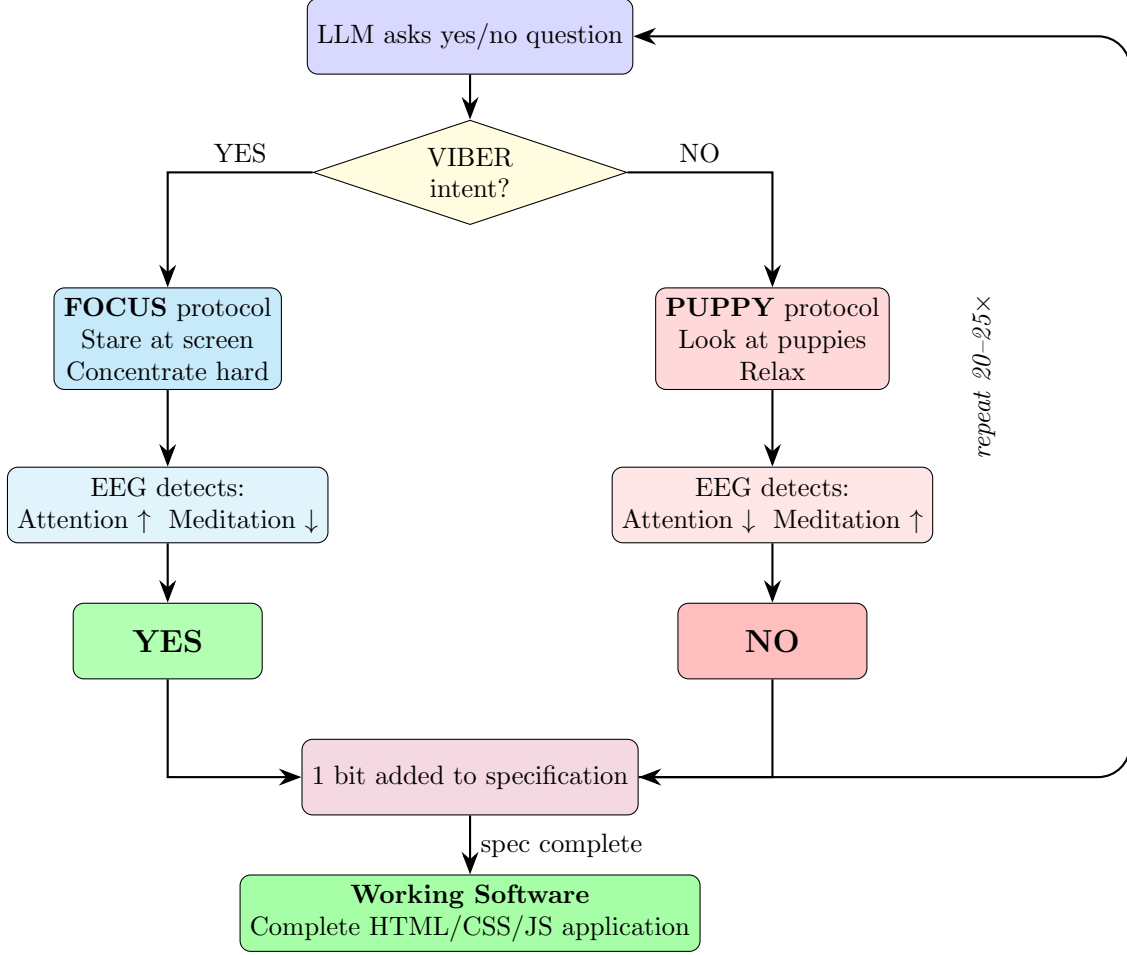


Figure 2: The BRAINROT decision loop. For each question, the VIBER (Volunteer for Intent-Based Electroencephalographic Research) either concentrates on the screen (FOCUS, indicating YES) or looks at pictures of puppies (PUPPY, indicating NO). The EEG headset reads the resulting attention/meditation shift and classifies it as a single bit. After 20–25 iterations, the LLM generates a complete application from the accumulated spec.

Binary decisions use differential thresholding: for each question, BRAINROT computes the change in Attention (ΔA) and Meditation (ΔM). $\Delta A > 0$ and $\Delta M < 0$ means YES. The inverse means NO. When both trend the same direction, the system waits.

3.2 The Questioning Engine

Given an initial prompt (“build me a web application”), the LLM generates yes/no questions to maximize information gain at each step (Figure 2).

The engine operates in two phases. In the **ex-**

ploration phase, questions are broad: “Is it a game?” “Does it involve managing a list?” Each answer eliminates roughly half the remaining possibility space. In the **refinement phase**, questions target specific features: “Dark theme?” “Local storage persistence?”

After 20–25 questions, the engine generates a complete, self-contained application implementing the accumulated spec. The LLM used was Claude (Anthropic).



Figure 3: The BRAINROT hardware platform: a NeuroSky TGAM EEG module (from a \$30 MindFlex toy) wired to a Raspberry Pi Zero 2W. State of the art.

3.3 FOCUS and PUPPY

FOCUS (Frontal Oscillation Concentration User Stimulus) is the YES protocol. The VIBER directs sustained attention to a fixed visual target, elevating the Attention metric relative to Meditation.

PUPPY (Parasympathetic Unwinding Protocol for Psychophysiological Yielding) is the NO protocol. The VIBER views a rotating selection of puppy images on a secondary display, activating parasympathetic relaxation pathways and elevating Meditation relative to Attention. The selection of canine imagery was motivated by established literature on the anxiolytic effects of animal observation [9], as well as the fact that the first author’s dog was present during early prototyping and appeared to measurably improve signal quality.

4 Methodology

We conducted two trials with a single subject (the first author, hereafter “the VIBER”). The VIBER wore the BRAINROT headset in a quiet room with two displays: one showing the questioning interface, one loaded with puppy images.

Both sessions began with the prompt “build me a web application.” The VIBER had a specific application in mind each time but communicated nothing to the system except binary neural

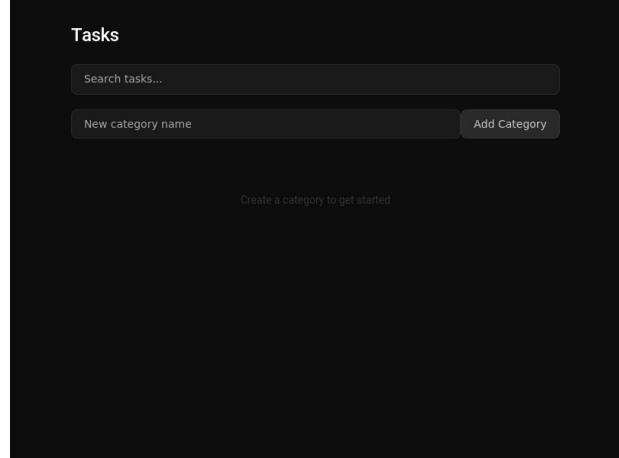


Figure 4: Session 1 output: a dark-themed task manager. 380 lines of HTML/CSS/JS from 24 binary brain signals.

responses. Session 1 targeted a task manager. Session 2 targeted a flashcard app.

For YES, the VIBER focused on the screen (typical attention delta: +10 to +50). For NO, the VIBER looked at puppies (typical meditation delta: +15 to +60). Each decision took 5–10 seconds. Total session time was 20–30 minutes per application.

5 Results

5.1 Session 1: Task Manager

24 questions to full specification. The engine identified “task manager” by question 5, then spent 19 questions on features. Output: a 380-line self-contained HTML application with dark theme, custom categories, search/filter, and local storage (Figure 4). It works.

Notable decisions: due dates (NO), drag-and-drop (NO), priority levels (NO), dark theme (YES, attention delta +46), animations (NO). The VIBER likes things minimal.

5.2 Session 2: Flashcard App

27 questions including one confirmation. The engine needed 15 questions just to identify the domain (Table 1), systematically eliminating games, dashboards, text editors, calculators,

#	Question	ΔA	Ans
1	Is this a web application?	+24	Y
2	Is this a game?	-20	N
3	Is it a productivity tool?	+42	Y
4	Does it involve a dashboard?	-31	N
5	Is it a text/writing tool?	-41	N
$\vdots$	(10 more NOs...)		
16	Does it involve learning?	+17	Y
17	Is it a flashcard/quiz tool?	+22	Y
18	Specifically flashcards?	+36	Y
19	User-created cards?	+49	Y
20	Multiple decks?	+19	Y
21	Spaced repetition?	+2	N
22	Score tracking?	+11	Y
23	Card flip animation?	+17	Y
24	Dark color scheme?	-23	N

Table 1: Excerpt from Session 2 (flashcard app). Questions 2–15 are a binary search over application categories. Once “learning” is identified at Q16, convergence to a full spec is rapid. ΔA = change in Attention metric.

task managers, timers, media tools, design tools, search tools, communication tools, generators, health apps, and finance apps before arriving at “learning/education” on question 16.

Once identified, convergence was rapid. Questions 16–27 specified: flashcards (not quiz), user-created cards, multiple decks, no spaced repetition, score tracking, flip animation, light theme, no persistence. Output: a 966-line HTML application. It also works.

The 15-question wander is a compelling visualization of binary search through possibility space, and an equally compelling demonstration that LLMs are not yet optimal at asking globally informative questions.

5.3 VIBER Self-Report

Metric	Score
“I felt like I was programming”	2/5
“I felt like a wizard”	5/5
“I trusted the output”	3/5
“The puppies helped”	5/5
“I would use this for production work”	1/5
“I would recommend this to a colleague”	4/5*

Table 2: Post-session VIBER satisfaction survey.

*For entertainment value.

6 Discussion

6.1 It Works, Which Is the Weird Part

Both sessions produced functional web applications. This raises an important question about programming: how much of it is specification versus implementation? If 24 binary decisions fully specify a working task manager, what are the other 47,000 keystrokes in a typical development session doing? The VIBER contributed intent. The LLM contributed everything else.

6.2 Binary Search over Possibility Space

Session 2’s first 15 questions were an $O(\log n)$ search over the space of all web applications. Each “no” eliminated a category, like a game of 20 Questions where the prize is a working code-base.

This suggests either that the possibility space of common web applications is smaller than it feels (a few hundred archetypes, navigable in ~ 8 bits), or that the LLM’s strong priors about “things people build” are doing most of the work and the brain signals are just steering. Probably both.

6.3 On Compression

We note in passing that 24 bits producing 380 lines of code is a compression ratio that would

make Claude Shannon reach for his whiskey. 27 bits yielding 966 lines is worse. We choose not to dwell on this, as its implications are either profound or meaningless, and we lack the philosophical training to determine which.

6.4 Limitations

Slow implementation speed. Compared to audio or keyboard based interfaces, BRAINROT produces working code considerably more slowly. Fixing this is trivial and left as an exercise to the reader. We just like, all got busy with like, life stuff you know?

7 Future Work

Telepathic pair programming. Two VIBERS, two headsets, one LLM. Preliminary analysis suggests adding a second neural signal would increase decision-making time by approximately 100%, consistent with existing pair programming literature.

Dream-driven CI/CD. REM-phase recordings triggering overnight build pipelines, enabling a literal “sleep on it” deployment strategy. The failure modes are concerning.

Cross-species generalization. During PUPPY development, the first author’s dog (Dr. Andi Dog, co-author) displayed sustained visual attention to the EEG dashboard, occasionally producing attention spikes when the signal quality indicator changed color. We do not claim this constitutes evidence of canine software engineering capability. We do not rule it out.

Scaling complexity. At ~40 lines per bit, generating a Linux kernel (~28 million lines) would require ~700,000 decisions, or roughly 4,000 years of continuous vibing. We consider this a reasonable timeline for kernel work.

Removing the human. If the LLM generates working code from 24 binary inputs, do those inputs need to come from a human? A random bit generator could theoretically produce functional software given enough attempts. For further work on random generation of software requirements, see the Penn MBA program.

8 Conclusion

We have presented BRAINROT, a system that generates working software from brain signals alone. It is extremely slow, mildly uncomfortable, and requires looking at pictures of puppies. In two trials, it produced functional web applications from approximately 25 binary decisions each.

This is the first time a complete software application has been generated using only a brain-computer interface as input. We encourage reviewers to evaluate this paper using BRAINROT. A single bit of feedback is, frankly, all most review comments contain anyway.

References

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